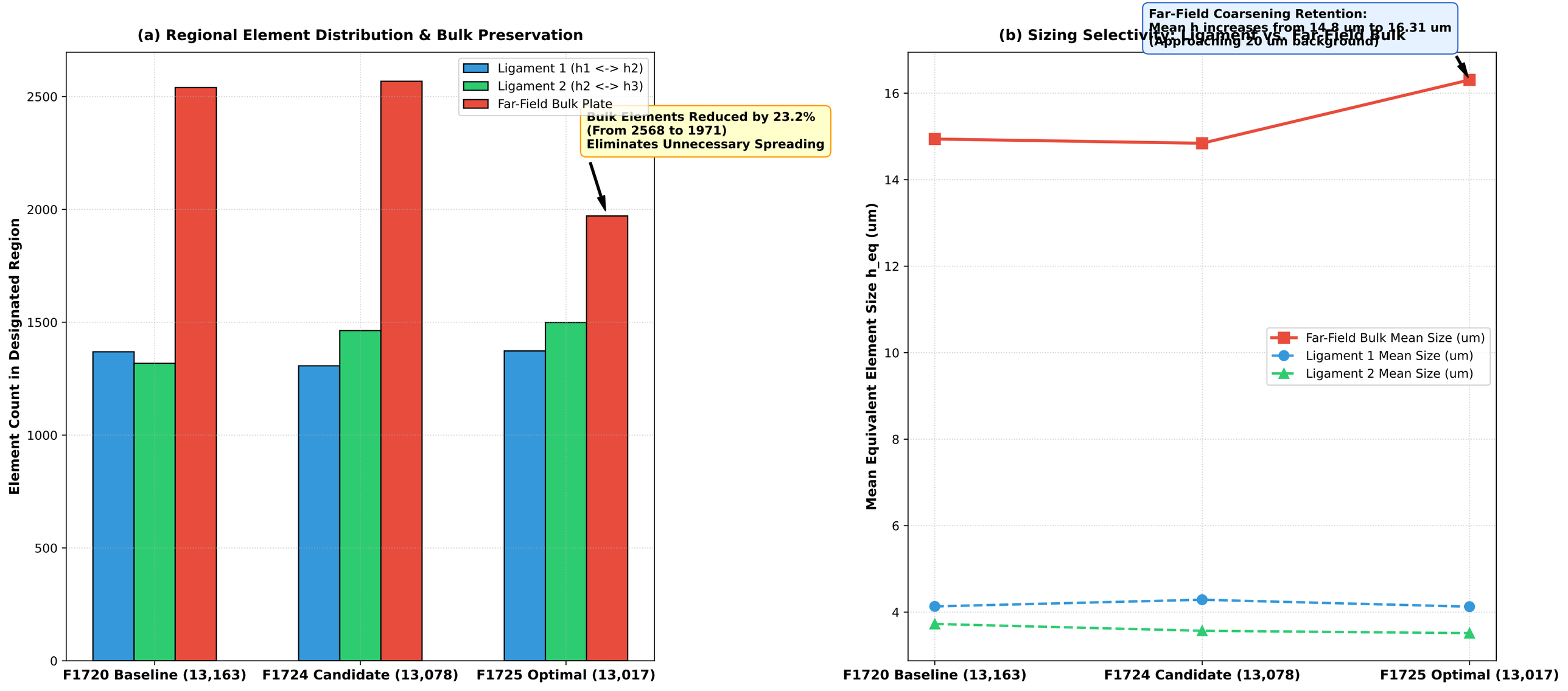
(a) Regional Element Distribution & Bulk Preservation
Element Count in Designated Region
Ligament 1 (h1 <-> h2)
Ligament 2 (h2 <-> h3)
Far-Field Bulk Plate
Bulk Elements Reduced by 23.2%
(From 2568 to 1971)
Eliminates Unnecessary Spreading
0
500
1000
1500
2000
2500
F1720 Baseline (13,163)
F1724 Candidate (13,078)
F1725 Optimal (13,017)
(b) Sizing Selectivity: Ligament vs. Far-Field Bulk
Mean Equivalent Element Size h_eq (um)
Far-Field Coarsening Retention:
Mean h increases from 14.8 um to 16.31 um
(Approaching 20 um background)
Far-Field Bulk Mean Size (um)
Ligament 1 Mean Size (um)
Ligament 2 Mean Size (um)
4
6
8
10
12
14
16
F1720 Baseline (13,163)
F1724 Candidate (13,078)
F1725 Optimal (13,017)